

11

Adaptations to Aerobic and Anaerobic Training

In this chapter and in the web study guide

Adaptations to Aerobic Training

Muscular Versus Cardiorespiratory Endurance

Evaluating Cardiorespiratory Endurance Capacity

Cardiovascular Adaptations to Training

Respiratory Adaptations to Training

Adaptations in Muscle

Metabolic Adaptations to Training

Integrated Adaptations to Chronic Endurance Exercise

What Limits Aerobic Power and Endurance Performance?

Long-Term Improvement in Aerobic Power and Cardiorespiratory Endurance

Factors Affecting an Individual's Response to Aerobic Training

Cardiorespiratory Endurance in Nonendurance Sports

Aerobic Deconditioning

VIDEO 11.1 presents Ben Levine on the significance of $\dot{V}O_{2\max}$ for sport performance.

AUDIO FOR FIGURE 11.7 describes how the variables of maximal cardiac output and red blood cell volume impact $\dot{V}O_{2\max}$ values in individuals.

AUDIO FOR FIGURE 11.8 describes the increases in total blood volume and plasma volume with endurance training.

ACTIVITY 11.1 *Adaptations* reviews the cardiovascular, respiratory, and metabolic responses to training.

AUDIO FOR FIGURE 11.15 describes a twin study on the effect of heredity on $\dot{V}O_{2\max}$.

ACTIVITY 11.2 *Individual Response* considers the factors affecting individual response to training.

ACTIVITY 11.3 *Aerobic Training* explores adaptations in response to aerobic training by applying them to real-life situations.

Adaptations to Anaerobic Training

Changes in Anaerobic Power and Anaerobic Capacity

Adaptations in Muscle with Anaerobic Training

Adaptations in the Energy Systems

ACTIVITY 11.4 *Anaerobic Training* explores adaptations in response to anaerobic training by applying them to real-life situations.

Adaptations to High-Intensity Interval Training

Specificity of Training and Cross-Training

In Closing

ACTIVITY 11.5 *Putting It All Together* reviews all concepts related to adaptations to aerobic and anaerobic training.



On October 8, 2016, the Ironman World Championships were held in Kona, on the Big Island of Hawaii, for the 40th time. Organized by the World Triathlon Corporation, professional triathletes qualified for the race based on a point system and a total of \$650,000 in prize money was awarded. German athlete Jan Frodeno completed this grueling event in 8:06:30 to win his second World Championship in as many years, completing the 3.9 km (2.4 mi) swim through tough ocean waves in just over 48 min, biking 180 km (112 mi) through hot lava fields in under 4.5 h, then running 42 km (26.2 mi) in 2:45:34. In the women's division, Daniela Ryf of Switzerland earned her second (and back-to-back) Ironman title, finishing almost 24 min ahead of the next closest woman competitor in 8:46:46—the sole woman's performance under 9 h. How are these athletes able to compete in this race? While there is little doubt that they are genetically gifted—including a high $\dot{V}O_{2\max}$ —rigorous sport-specific training is also required to develop their cardiorespiratory endurance capacities.

During a single bout of aerobic exercise, the human body precisely adjusts its cardiovascular and respiratory function to meet the energy and oxygen demands of actively contracting muscle. When these systems are challenged repeatedly, as happens with regular exercise training, they adapt in ways that allow the body to improve $\dot{V}O_{2\max}$ and overall endurance performance. **Aerobic training**, or cardiorespiratory endurance training, improves cardiac function and peripheral blood flow and enhances the capacity of the muscle fibers to generate greater amounts of adenosine triphosphate (ATP). In this chapter, we examine adaptations in cardiorespiratory function in response to endurance training and how such adaptations improve an athlete's endurance capacity and performance. Additionally, we examine adaptations to anaerobic training. **Anaerobic training** improves anaerobic metabolism; short-term, high-intensity exercise capacity; tolerance for acid–base imbalances; and in some cases, muscle strength. Both aerobic and anaerobic training induce a variety of adaptations that benefit exercise and sport performance.

The effects of training on cardiovascular and respiratory, or aerobic, endurance are well known to endurance athletes like distance runners, cyclists, cross-country skiers, and swimmers but

are often ignored by other types of athletes. Training programs for many nonendurance athletes often ignore the aerobic endurance component. This is understandable, because for maximum improvement in performance, training should be highly specific to the particular sport or activity in which the athlete participates, and endurance is frequently not recognized as important to nonendurance activities. The reasoning is, why waste valuable training time if the result is not improved performance?

The problem with this reasoning is that most nonendurance sports do indeed have an endurance, or aerobic, component. For example, in football, players and coaches might fail to recognize the importance of cardiorespiratory endurance as part of the total training program. From all outward appearances, American football is an anaerobic, or burst-type, activity consisting of repeated bouts of high-intensity work of short duration. Seldom does a run exceed 40 to 60 yd (37-55 m), and even this is usually followed by a substantial rest interval. The need for endurance may not be readily apparent. What athletes and coaches might fail to consider is that this burst-type activity must be repeated many times during the game. With a higher aerobic endurance capacity, an athlete could maintain the quality of each burst activity throughout the game and would still be relatively fresh (less drop-off in performance, fewer feelings of fatigue) during the all-important closing minutes of the game.

Chapters 9 and 14 cover the principles of training for sport performance—the “how,” “when,” and “how much” questions about how training improves athletic performance. The focus here is on those physiological changes that occur within the body systems when aerobic or anaerobic exercise is repeated regularly to induce a training response.

Adaptations to Aerobic Training

Improvements in endurance that accompany regular (e.g., daily, every other day) aerobic training, such as running, cycling, or swimming, result from multiple adaptations to the training stimuli. Some adaptations occur in the cardiovascular system, improving circulation to and within the muscles. Still other important changes occur within the muscles themselves, promoting more efficient

utilization of oxygen and fuel substrates. Pulmonary adaptations, as will be noted later, occur to a lesser extent.

Muscular Versus Cardiorespiratory Endurance

Endurance is a term that refers to two separate but related concepts: muscular endurance and cardiorespiratory endurance. Each makes a unique contribution to athletic performance, and each differs in its importance to different athletes.

For sprinters, endurance is the quality that allows them to sustain a high speed over the full distance of, for example, a 100 m or 200 m dash. This component of fitness is termed muscular endurance, the ability of a single muscle or muscle group to maintain high-intensity, repetitive, or static contractions. This type of endurance is also exemplified by a weightlifter doing multiple repetitions, a boxer, or a wrestler. The exercise or activity can be rhythmic and repetitive in nature, such as multiple repetitions of the bench press for the weightlifter and jabbing for the boxer. Or the activity can be more static, such as a sustained muscle action when a wrestler attempts to pin an opponent. In either case, the resulting fatigue is confined to a specific muscle group, and the activity's duration is usually no more than 2 min. Muscular endurance is highly related to muscular strength and to anaerobic power development.

While muscular endurance is specific to individual muscles or muscle groups, cardiorespiratory endurance relates to the ability to sustain prolonged, dynamic whole-body exercise using large muscle groups. Cardiorespiratory endurance is related to the development of the cardiorespiratory systems' ability to maintain oxygen delivery to working muscles during prolonged exercise, as well as the muscles' ability to use energy aerobically (discussed in [chapters 2](#) and [5](#)). This is why the terms *cardiorespiratory endurance* and *aerobic endurance* are sometimes used synonymously.

Evaluating Cardiorespiratory Endurance Capacity

Studying the effects of training on endurance requires an objective, repeatable means of measuring an individual's cardiorespiratory endurance capacity. In that way, the exercise scientist, coach, or athlete can monitor improvements as physiological adaptations occur during the training program.

Maximal Endurance Capacity: $\dot{V}O_{2max}$

Most exercise scientists regard $\dot{V}O_{2max}$, sometimes called maximal aerobic power or maximal aerobic capacity, as the best objective laboratory measure of cardiorespiratory endurance. Recall from [chapter 5](#) that $\dot{V}O_{2max}$ is defined as the highest rate of oxygen consumption attainable during maximal or exhaustive exercise. $\dot{V}O_{2max}$ as defined by the Fick equation is determined by maximal cardiac output (delivery of oxygen and blood flow to working muscles) and the maximal $(a-\bar{v})O_2$ difference (the ability of the active muscles to extract and use the oxygen).

As exercise intensity increases, oxygen consumption eventually either plateaus or decreases slightly, even with further increases in workload, indicating that a true maximal $\dot{V}O_{2max}$ has been achieved.

With endurance training, more oxygen can be delivered to, and used by, active muscles than in an untrained state. Previously untrained subjects demonstrate average increases in $\dot{V}O_{2max}$ of 15% to 20% after a 20-week training program. These improvements allow individuals to perform endurance activities at a higher intensity, improving their performance potential. [Figure 11.1](#) illustrates the increase in $\dot{V}O_{2max}$ after 12 months of aerobic training in a previously untrained individual. In this example, $\dot{V}O_{2max}$ increased by about 30%. Note that the $\dot{V}O_2$ cost of running at a certain submaximal intensity (referred to as running economy) did not change, but higher running speeds could be attained after training.

VIDEO 11.1 Presents Ben Levine on the significance of $\dot{V}O_{2max}$ for sport performance.





Submaximal Endurance

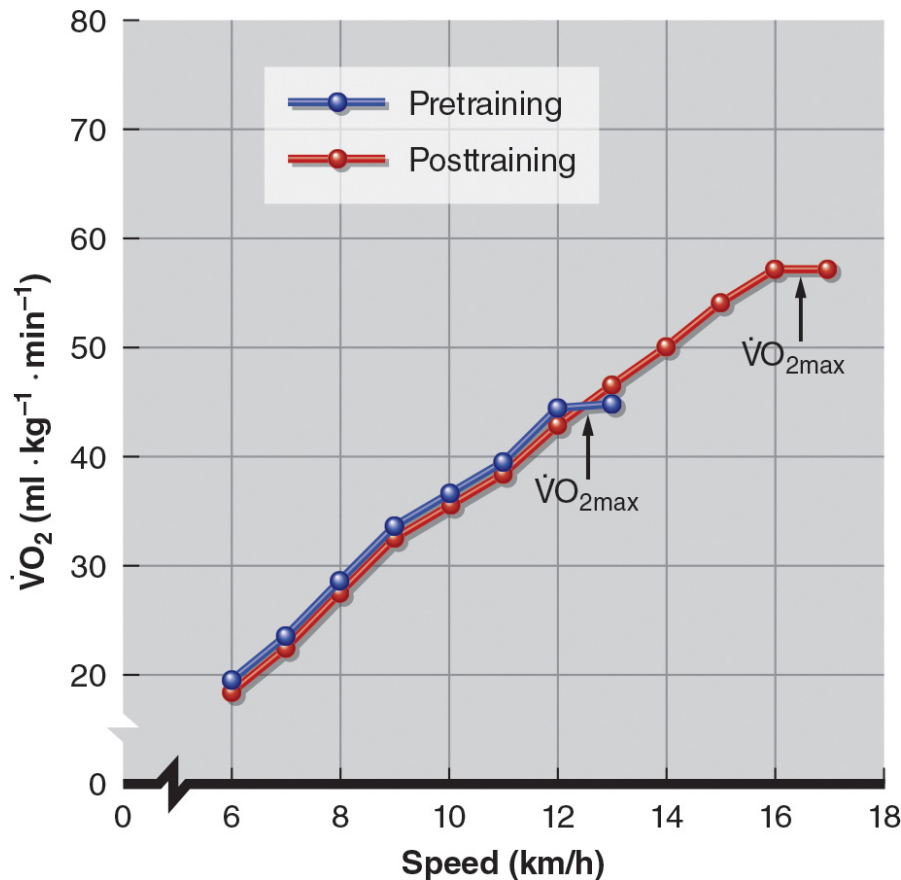


FIGURE 11.1 Changes in $\dot{V}O_{2\max}$ with 12 months of endurance training. $\dot{V}O_{2\max}$ increased from 44 to 57 ml · kg⁻¹ · min⁻¹, a 30% increase. Peak speed during the treadmill test increased from 13 km/h (8 mph) to 16 km/h (~10 mph).

In addition to increasing maximal endurance capacity, endurance training also increases **submaximal endurance**, which is more difficult to evaluate. A lower steady-state heart rate at the same submaximal exercise intensity is one physiological variable that can be used to objectively quantify the effect of training. Additionally, one could measure the average peak absolute power output a person can maintain over a fixed period of time on a cycle ergometer. For running, the average peak speed or velocity a person can maintain for a set period of time would be a similar test of submaximal

endurance. Generally, these tests last 30 min to an hour and reflect the concept of critical power discussed in [chapter 5](#). Submaximal endurance, like critical power, is more closely related to actual competitive endurance performance than $\dot{V}O_{2\max}$. With endurance training, submaximal endurance increases.

RESEARCH PERSPECTIVE 11.1

How Much Can $\dot{V}O_{2\max}$ Improve?

In 1968, the Dallas bed-rest study demonstrated that $\dot{V}O_{2\max}$ could be doubled (from roughly $25 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ to $50 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) within a few weeks of training after a period of detraining.³⁰ Despite this huge increase in $\dot{V}O_{2\max}$ following bed-rest induced detraining, $50 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ is a rather typical $\dot{V}O_{2\max}$ for a recreational endurance athlete, and it is unlikely that an average active adult can increase $\dot{V}O_{2\max}$ from this average to values even remotely close to double.²¹ Meanwhile, elite endurance athletes typically have $\dot{V}O_{2\max}$ values approaching $80 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, with the highest value ever published at an incredible $90.6 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ in an Olympic gold medalist cross-country skier. It is unlikely that any ordinary human can achieve such astounding values even with rigorous training programs, so how, and by how much, can $\dot{V}O_{2\max}$ actually be enhanced?

Large $\dot{V}O_{2\max}$ changes may take years to achieve. Prospective training studies are challenging to undertake in the laboratory, but the longest published study showed only a 21% increase in $\dot{V}O_{2\max}$ over 12 months of training at moderate to high intensities every other day. Other training studies lasting 4 to 6 months show even more modest increases of 9% to 17%, and overall it appears that average endurance training improves $\dot{V}O_{2\max}$ $\sim 0.5 \text{ L/min}$. High-intensity interval training has shown the largest increase in $\dot{V}O_{2\max}$ (44%), but it should be noted that the training intensities and volumes in all of these studies are far lower than the training load of world-class athletes. In contrast to the average-fit subjects in these longitudinal studies, young (15-25 years of age) world-class athletes can substantially increase their $\dot{V}O_{2\max}$ from an already high 55 to $60 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ to 75 to $80 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ with years of intense training. This phenomenon suggests that large increases in $\dot{V}O_{2\max}$ can be seen in athletes undergoing intense training and that training in early life is likely a determinant of very high $\dot{V}O_{2\max}$ values recorded in endurance champions.

A high $\dot{V}O_{2\max}$ is the product of a high maximal cardiac output ($\dot{Q}_{\max}$) and a high oxygen-carrying capacity of the blood. This maximal blood flow and

oxygen carrying leads to increased oxygen delivery to the exercising muscles. The high $\dot{Q}_{\max}$ in elite athletes is the result of increased stroke volume, since maximal heart rate does not change with training. Endurance trained athletes achieve this higher stroke volume through changes in the left ventricle of the heart such that it has a larger mass and is more distensible and therefore easier to fill with each heartbeat. But what about nonelite athletes? It remains unknown if average individuals can ever reach $\dot{Q}_{\max}$ values observed in elite athletes. The initial increase in stroke volume observed with training healthy, normal volunteers is due to an increased blood volume rather than changes in the myocardium. After 1 year of exercise training in previously untrained individuals, left ventricle mass has been shown to increase. However, these changes do not result in very large increases in $\dot{Q}_{\max}$. These findings suggest that it is unlikely that individuals with average cardiac function can ever reach values observed in elite athletes, but it may be that exercise training during childhood and early adulthood may favor the development of these advantageous cardiac characteristics.

Given the limited potential for training to increase $\dot{Q}_{\max}$, the adaptation for improved oxygen delivery is an important factor for the improvement of $\dot{V}O_{2\max}$ with endurance training. Oxygen-carrying capacity is directly related to the number of hemoglobin available to bind to oxygen, and hemoglobin mass correlates tightly with exercise performance. There is little doubt that exercise training increases hemoglobin mass or total red blood cell volume ~20%. It is unknown if long-term endurance training can increase hemoglobin mass from normal values (~700 g) to that observed in elite athletes (~1,200 g), but it appears unlikely. There may be genetic determinants of total hemoglobin mass, but research has thus far failed to find a genetic polymorphism to explain extremely high hemoglobin mass in elite athletes.

Finally, improvements in oxygen extraction, i.e., increases in (a-v) O_2 difference, may also contribute to the increase in $\dot{V}O_{2\max}$ with training. Athletes have a more homogeneous blood flow distribution during submaximal exercise, which results in a higher oxygen extraction compared to untrained individuals. Systemic maximal oxygen extraction can be improved with training in healthy volunteers, from 72% up to 84%. Although this is a meaningful improvement, it is still nowhere near oxygen extraction reported in elite endurance athletes (93%). It is unlikely that average individuals can achieve the high oxygen extractions observed in elite athletes; however, the possibility that already trained elite endurance athletes can further improve oxygen extraction has yet to be studied.

Overall, although endurance training leads to improvements in the mechanisms contributing to $\dot{V}O_{2\max}$, the overall increases observed in healthy, normal individuals rarely exceed 0.5 L/min and never reach the extraordinarily high values observed in elite endurance athletes. Even highly trained athletes seem to plateau after age 25, and increases in performance after that are due to increases in other mechanisms such as mechanical efficiency or critical

power. $\dot{V}O_{2\max}$ is a powerful determinant of endurance performance, but the magnitude of improvements that can be achieved through training are relatively small, even in elite endurance athletes.²⁰

Cardiovascular Adaptations to Training

Multiple cardiovascular adaptations occur in response to exercise training, including changes in the following:

- Heart size
- Stroke volume
- Heart rate
- Cardiac output
- Blood flow
- Blood and red cell volumes

Not surprisingly, these variables are interrelated. For example, training-induced increases in stroke volume depend on increases in both heart size and blood volume. To fully understand adaptations in these variables, it is important to review how these components relate to oxygen transport.

Oxygen Transport System

Cardiorespiratory endurance is related to the cardiovascular and respiratory systems' ability to deliver sufficient oxygen to meet the needs of metabolically active tissues.

Recall from [chapter 8](#) that the ability of the cardiovascular and respiratory systems to deliver oxygen to active tissues is defined by the **Fick equation**, which states that whole-body oxygen consumption is determined by both the delivery of oxygen via blood flow (cardiac output) and the amount of oxygen extracted by the tissues, the $(a-\bar{v})O_2$ difference. The product of cardiac output and the $(a-\bar{v})O_2$ difference determines the rate at which oxygen is being consumed:

$$\dot{V}O_2 = \text{stroke volume} \times \text{heart rate} \times (a-\bar{v})O_2 \text{ difference}$$

and

$$\dot{V}O_{2\max} = \text{maximal stroke volume} \times \text{maximal heart rate} \times \text{maximal } (a-\bar{v})O_2 \text{ difference}$$

Because maximal heart rate ($HR_{\max}$) either stays the same or decreases slightly with training, increases in $\dot{V}O_{2\max}$ depend on adaptations in maximal stroke volume and maximal $(a-\bar{v})O_2$ difference.

The oxygen demand of exercising muscles increases with increasing exercise intensity. Aerobic endurance depends on the cardiorespiratory system's ability to deliver sufficient oxygen to these active tissues to meet their heightened demands for oxygen for oxidative metabolism. As maximal levels of exercise are achieved, heart size, blood flow, blood pressure, and blood volume can all potentially limit the maximal ability to transport oxygen. Endurance training elicits numerous changes in these components of the **oxygen transport system** that enable it to function more effectively.

Heart Size

The measurement of heart size has been of interest to cardiologists for years because a hypertrophied, or enlarged, heart is typically a pathological condition indicating the presence of cardiovascular disease. Clinicians and scientists commonly use echocardiography to accurately measure the size of the heart and its chambers. Echocardiography involves the technique of ultrasound, which uses high-frequency sound waves directed through the chest wall to the heart. These sound waves are emitted from a transducer placed on the chest; once they contact the various structures of the heart, they rebound back to a sensor, which is able to capture the deflected sound waves and provide a moving picture of the heart. A trained physician or technician can visualize the size of the heart's chambers, thicknesses of its walls, and heart valve action. There are several forms of echocardiography: M-mode echocardiography, which provides a one-dimensional view of the heart; two-dimensional echocardiography; and Doppler echocardiography, which is used more often to measure blood flow through large arteries.

As an adaptation to the increased work demand, cardiac muscle mass and ventricular volume increase with training. Cardiac muscle, like skeletal muscle, undergoes morphological adaptations as a result

of chronic endurance training. At one time, **cardiac hypertrophy** induced by exercise—**athlete's heart**, as it was called—was viewed with concern because experts incorrectly believed that enlargement of the heart always reflected a pathological state, as sometimes occurs with severe hypertension. Training-induced cardiac hypertrophy, on the other hand, is now recognized as a normal adaptation to chronic endurance training.

The left ventricle, as discussed in [chapter 6](#), does the most work and thus undergoes the greatest adaptation in response to endurance training. The type of ventricular adaptation depends on the type of exercise training performed. For example, during resistance training, the left ventricle must contract against increased afterload from the systemic circulation. From [chapter 8](#) we learned that blood pressure during resistance exercise can exceed 480/350 mmHg. This presents a considerable resistance that must be overcome by the left ventricle. To overcome this high afterload, the heart muscle compensates by increasing left ventricular wall thickness, thereby increasing its contractility. Thus, the increase in its muscle mass is in direct response to repeated exposure to the increased afterload with resistance training. However, there is little change in ventricular volume.

With endurance training, left ventricular chamber size increases. This allows for increased left ventricular filling and consequently an increase in stroke volume. The increase in left ventricular dimensions is largely attributable to a training-induced increase in plasma volume (discussed later in this chapter) that increases left ventricular end-diastolic volume (increased preload). In concert with this, a decrease in heart rate at rest caused by increased parasympathetic tone, and during exercise at the same rate of work, allows a longer diastolic filling period. The increases in plasma volume and diastolic filling time increase left ventricular chamber size at the end of diastole. This effect of endurance training on the left ventricle is often called a volume loading effect.

It was originally hypothesized that this increase in left ventricular dimensions was the only change in the left ventricle caused by endurance training. Additional research has revealed that, similar to what happens in resistance training, myocardial wall thickness

increases with endurance training. Highly trained endurance athletes (competitive cross-country skiers, endurance cyclists, and long-distance runners) have greater left ventricular masses than non-endurance-trained men and women. Furthermore, left ventricular mass is highly correlated with $\dot{V}O_{2\max}$.

Fagard¹² conducted the most extensive review of the existing research literature in 1996, focusing on long-distance runners (135 athletes and 173 controls), cyclists (69 athletes and 65 controls), and strength athletes (178 athletes, including weight- and powerlifters, bodybuilders, wrestlers, throwers, and bobsledders, and 105 controls). For each group, the athletes were matched by age and body size with a group of sedentary control subjects. For each group of runners, cyclists, and strength athletes, the internal diameter of the left ventricle (LVID, an index of chamber size) and the total left ventricular mass (LVM) were greater in the athletes compared with their age- and sized-matched controls (figure 11.2). Thus, data from this review support the hypothesis that both left ventricular chamber size and wall thickness increase with endurance training.

Most studies of heart size changes with training have been cross-sectional, comparing trained individuals with sedentary, untrained individuals. Certainly a portion of the differences that we see in figure 11.2 can be attributed to genetics, not training. However, a number of longitudinal studies have followed individuals from an untrained state to a trained state, and others have followed individuals from a trained state to an untrained state. These studies have reported increases in heart size with training and decreases with detraining. Therefore, training does bring about changes, but they are likely not as large as the differences shown in figure 11.2.

In Review

- *Cardiorespiratory endurance* (also called *maximal aerobic power*) refers to the ability to perform prolonged, dynamic exercise using a large muscle mass.
- $\dot{V}O_{2\max}$ —the highest rate of oxygen consumption obtainable during maximal or exhaustive exercise—is the best single measure of cardiorespiratory endurance.
- Cardiac output, the product of heart rate and stroke volume, represents how much blood leaves the heart each minute, whereas $(a-\bar{v})O_2$ difference is a measure of how much oxygen is extracted from the blood by the tissues. According to the

Fick equation, the product of these values is the rate of oxygen consumption: $\dot{V}O_2$ = stroke volume $\times$ heart rate $\times$ (a- $\bar{V}O_2$ difference).

- Of the chambers of the heart, the left ventricle adapts the most in response to endurance training.
 - With endurance training, the internal dimensions of the left ventricle increase, mostly in response to an increase in ventricular filling secondary to an increase in plasma volume.
 - Left ventricular wall thickness and mass also increase with endurance training, allowing for a greater force of contraction.
-

Stroke Volume

Stroke volume at rest is substantially higher after an endurance training program than it is before training. This endurance training-induced increase is also seen at a given submaximal exercise intensity and at maximal exercise. This increase is illustrated in [figure 11.3](#), which shows the changes in stroke volume of a subject who exercised at increasing intensities up to a maximal intensity before and after a 6-month endurance training program. Typical values for stroke volume at rest and during maximal exercise in untrained, trained, and highly trained athletes are listed in [table 11.1](#). The wide range of stroke volume values for any given cell within this table is largely attributable to differences in body size. Larger people typically have larger hearts and a greater blood volume, and thus higher stroke volumes—an important point when one is comparing stroke volumes of different people.

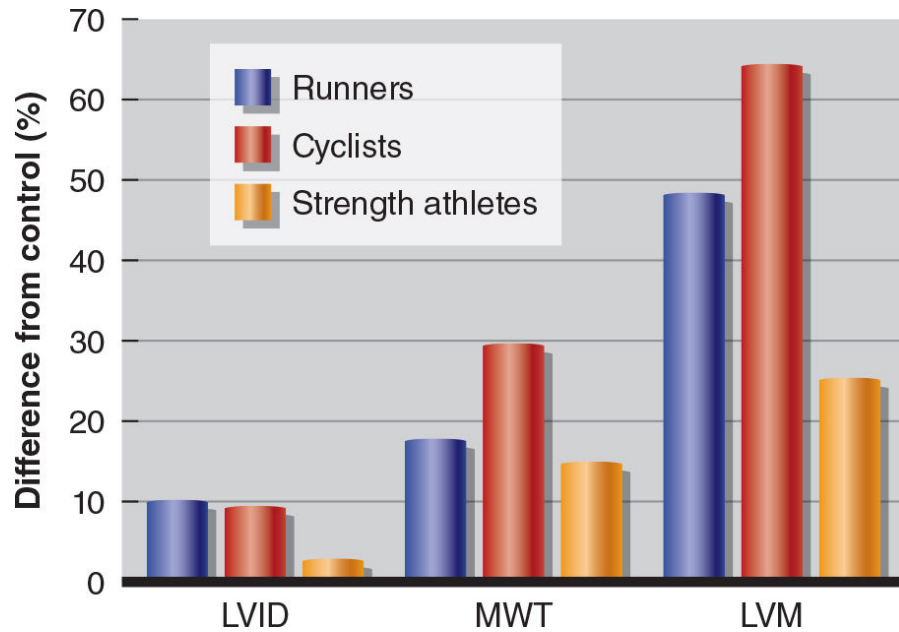


FIGURE 11.2 Percentage differences in heart size of three groups of athletes (runners, cyclists, and strength athletes) compared with their age- and size-matched sedentary controls (0%). Percentage differences are presented for left ventricular internal diameter (LVID), mean wall thickness (MWT), and left ventricular mass (LVM).

Data are from Fagard (1996).

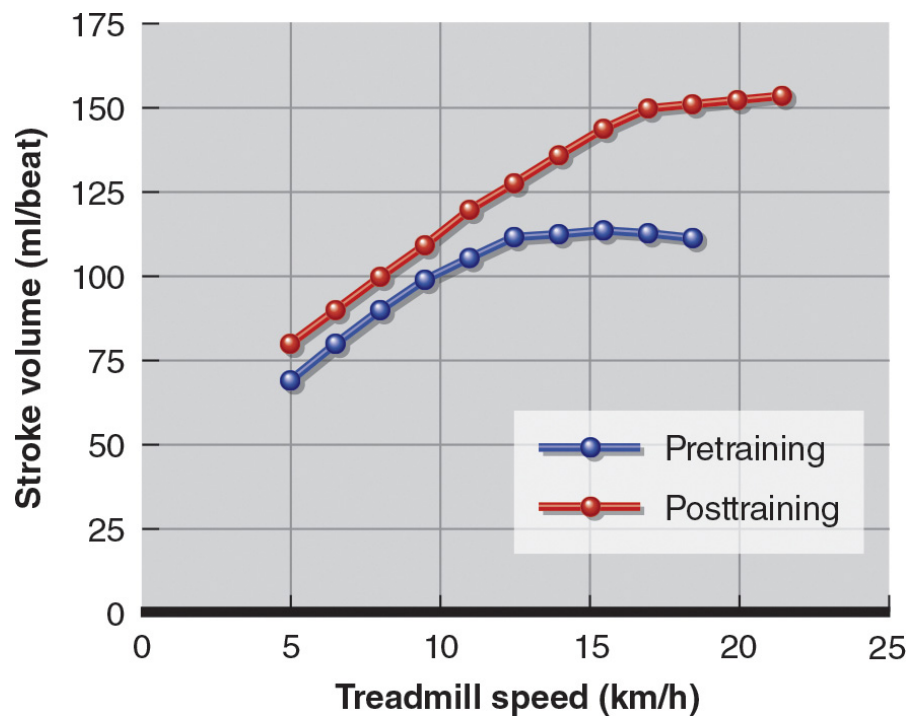


FIGURE 11.3 Changes in stroke volume with endurance training during walking, jogging, and running on a treadmill at increasing velocities.

TABLE 11.1 Stroke Volumes at Rest (SV_{rest}) and During Maximal Exercise (SV_{max}) for Different States of Training

Subjects	SV_{rest} (ml/beat)	SV_{max} (ml/beat)
Untrained	50-70	80-110
Trained	70-90	110-150
Highly trained	90-110	150-220+

After aerobic training, the left ventricle fills more completely during diastole. Plasma volume expands with training, which allows for more blood to enter the ventricle during diastole, increasing end-diastolic volume (EDV). The heart rate of a trained heart is also lower at rest and at the same absolute exercise intensity than that of an untrained heart, allowing more time for the increased diastolic filling. More blood entering the ventricle increases the stretch on the ventricular walls; by the Frank-Starling mechanism (see [chapter 8](#)), this results in an increased force of contraction.

The thickness of the posterior and septal walls of the left ventricle also increases slightly with endurance training. Increased ventricular muscle mass results in increased contractile force, in turn causing a lower end-systolic volume (ESV).

The decrease in ESV is facilitated by the decrease in peripheral resistance that occurs with training. Increased contractility resulting from an increase in left ventricular thickness and greater diastolic filling (Frank-Starling mechanism), coupled with the reduction in systemic peripheral resistance, increases the ejection fraction [equal to $(EDV - ESV) / EDV$] in the trained heart. More blood enters the left ventricle, and a greater percentage of what enters is forced out with each contraction, resulting in an increase in stroke volume.

Adaptations in stroke volume during endurance training are illustrated by a study in which older men trained aerobically for 1 year.¹⁰ Their cardiovascular function was evaluated before and after training. The subjects performed running, treadmill, and cycle ergometer exercise for 1 h each day, 4 days per week. They exercised at intensities of 60% to 80% of $\dot{V}O_{2max}$, with brief bouts of exercise exceeding 90% of $\dot{V}O_{2max}$. End-diastolic volume increased at rest and throughout submaximal exercise. The ejection fraction increased, which was associated with a decreased ESV, suggesting

increased contractility of the left ventricle. $\dot{V}O_{2\max}$ increased by 23%, a substantial improvement in endurance.

To summarize, increased left ventricular dimensions, reduced systemic peripheral resistance, and a greater blood volume account for the increases in resting, submaximal, and maximal stroke volume after an endurance training program.

In Review

- Following endurance training, stroke volume (SV) is increased at rest and during submaximal and maximal exercise.
- A major factor leading to the SV increase is an increased end-diastolic volume (EDV) caused by an increase in plasma volume and a greater diastolic filling time secondary to a lower heart rate.
- Another contributing factor to increased SV is an increased left ventricular force of contraction. This is caused by hypertrophy of the cardiac muscle and increased ventricular stretch resulting from an increase in diastolic filling (increased preload), leading to greater elastic recoil (Frank-Starling mechanism).
- Reduced systemic vascular resistance (decreased afterload) also contributes to the increased volume of blood pumped from the left ventricle with each beat.

Heart Rate

Aerobic training has a major impact on heart rate at rest, during submaximal exercise, and during the postexercise recovery period. The effect of aerobic training on maximal heart rate is rather negligible.

Resting Heart Rate Resting heart rate decreases markedly as a result of endurance training. Some studies have shown that a sedentary individual with an initial resting heart rate of 80 beats/min can decrease resting heart rate by approximately 1 beat/min with each week of aerobic training, at least for the first few weeks. After 10 weeks of moderate endurance training, resting heart rate can decrease from 80 to 70 beats/min or lower. On the other hand, well-controlled studies with large numbers of subjects have shown much smaller decreases in resting heart rate, that is, fewer than 5 beats/min following up to 20 weeks of aerobic training.

Recall from [chapter 6](#) that bradycardia is a term indicating a heart rate of fewer than 60 beats/min. In untrained individuals, bradycardia can be the result of abnormal cardiac function or heart disease. However, highly conditioned endurance athletes often have resting heart rates lower than 40 beats/min, and some have values lower than 30 beats/min. Therefore, it is necessary to differentiate between training-induced bradycardia, which is a normal response to endurance training, and pathological bradycardia, which can be cause for concern.

The low resting heart rate (HR) of well-trained endurance athletes is most often attributed to an elevated parasympathetic (vagal) tone. However, a 2013 review of the available evidence casts doubt on this mechanism.⁶ The two alternative explanations for the resting bradycardia of athletes are a diminished sympathetic tone and a lower intrinsic heart rate. Recall from [chapter 6](#) that the intrinsic heart rate is the rate of sinoatrial (SA) node firing in the absence of any neural or hormonal input. In studies that have blocked parasympathetic activity to the heart using the drug atropine, there is still a significant resting bradycardia in athletes. In fact, the difference in HR after parasympathetic blockade is greater than the difference in the normal HR, suggesting that the bradycardia is not the result of elevated vagal tone.

Other studies have blocked both branches of the autonomic nervous system, that is, used a complete autonomic blockade. The HR after complete autonomic blockade is a measure of the intrinsic HR. In studies showing a lowered resting HR after endurance training, the bradycardia persists after complete autonomic blockade. Thus, the resting bradycardia seen in athletes is at least partially, and perhaps completely, the result of a decreased intrinsic HR.

A decreased intrinsic HR can result from a remodeling of the SA node. The SA node serves as the pacemaker of the heart due to properties of ion channels and Ca^{2+} -handling proteins in the SA node cells. Changes in these properties cause the well-known bradycardias associated with SA node disease, heart failure, atrial fibrillation, and even aging itself. In fact, the age-associated decrease in resting HR has been attributed to a downregulation of ryanodine receptors (see [chapter 6](#)) that are involved in Ca^{2+} flux. If these

mechanisms are involved in bradycardias associated with these other processes and diseases, it is likely that they are involved in training-induced bradycardia as well.

Submaximal Heart Rate During submaximal exercise, aerobic training results in a lower heart rate at any given absolute exercise intensity. This is illustrated in [figure 11.4](#), which shows the heart rate of an individual exercising on a treadmill before and after training. At each walking or running speed, the posttraining heart rate is lower than the heart rate before training. The training-induced decrease in heart rate is typically greater at higher intensities.

While maintaining a cardiac output appropriate to meet the needs of working muscle, a trained heart performs less work (lower heart rate, higher stroke volume) than an untrained heart at the same absolute workload.

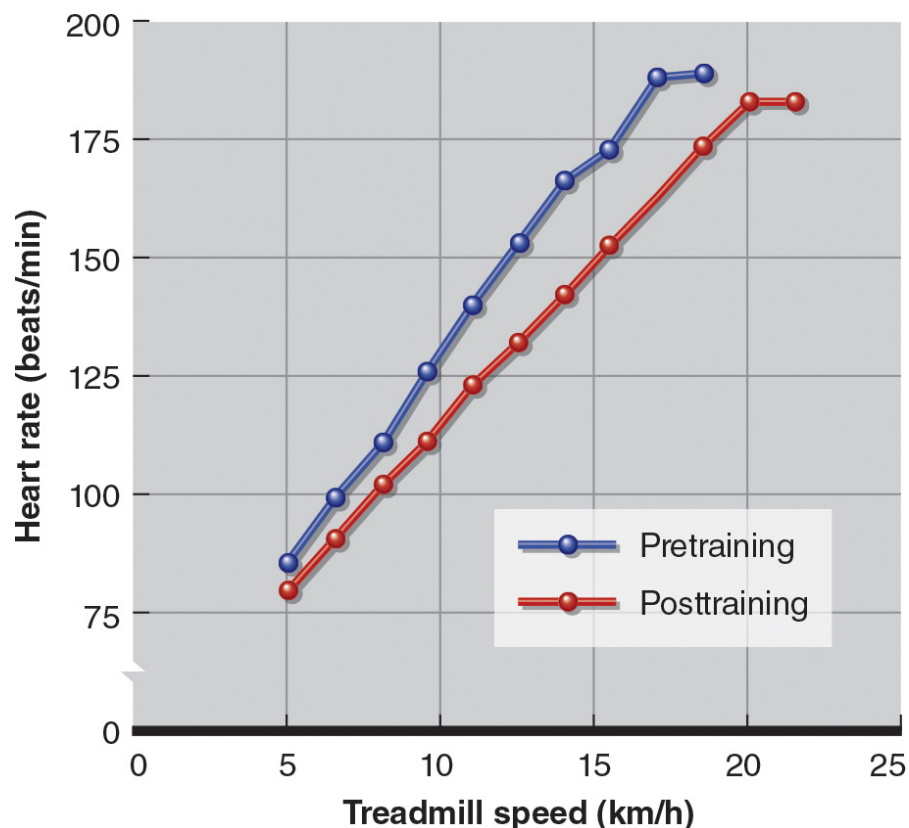


FIGURE 11.4 Endurance training-induced changes in heart rate during progressive walking, jogging, and running on a treadmill at increasing speeds.

Maximum Heart Rate A person's maximal heart rate (HR_{max}) tends to be stable and typically remains relatively unchanged after endurance training. However, several studies have suggested that for people whose untrained HR_{max} values exceed 180 beats/min, HR_{max} might be slightly lower after training. Also, highly conditioned endurance athletes often have lower HR_{max} values than untrained individuals of the same age, although this is not always the case. Athletes over 60 years old sometimes have higher HR_{max} values than untrained people of the same age.

Interactions Between Heart Rate and Stroke Volume During exercise, the product of heart rate and stroke volume provides a cardiac output appropriate to the intensity of the activity being performed. At maximal or near-maximal intensities, heart rate may change to provide the optimal combination of heart rate and stroke volume to maximize cardiac output. If heart rate is too fast, diastolic filling time is reduced, and stroke volume might be compromised. For example, if HR_{max} is 180 beats/min, the heart beats three times per second. Each cardiac cycle thus lasts for only 0.33 s. Diastole is as short as 0.15 s or less. This fast heart rate allows very little time for the ventricles to fill. As a consequence, stroke volume may decrease at high heart rates when filling time is compromised.

However, if the heart rate slows, the ventricles have longer to fill. This has been proposed as one reason highly trained endurance athletes tend to have lower HR_{max} values: Their hearts have adapted to training by drastically increasing their stroke volumes, so lower HR_{max} values can provide optimal cardiac output.

Which comes first? Does increased stroke volume result in a decreased heart rate, or does a lower heart rate result in an increased stroke volume? This question remains unanswered. In either case, the combination of increased stroke volume and decreased heart rate is a more efficient way for the heart to meet the metabolic demands of the exercising body. The heart expends less energy by contracting less often but more forcefully than it would if contraction frequency were increased. Reciprocal changes in heart rate and stroke volume in response to training share a common goal: to allow the heart to pump the maximal amount of oxygenated blood at the lowest energy cost.

Heart Rate Recovery During exercise, as discussed in [chapter 6](#), heart rate must increase to increase cardiac output to meet the blood flow demands of active muscles. When the exercise bout is finished, heart rate does not instantly return to its resting level. Instead, it remains elevated for a while, slowly returning to its resting rate. The time it takes for heart rate to return to its resting rate is called the heart rate recovery period.

After endurance training, as shown in [figure 11.5](#), heart rate returns to its resting level much more quickly after an exercise bout than it does before training. This is true after both submaximal and maximal exercise.

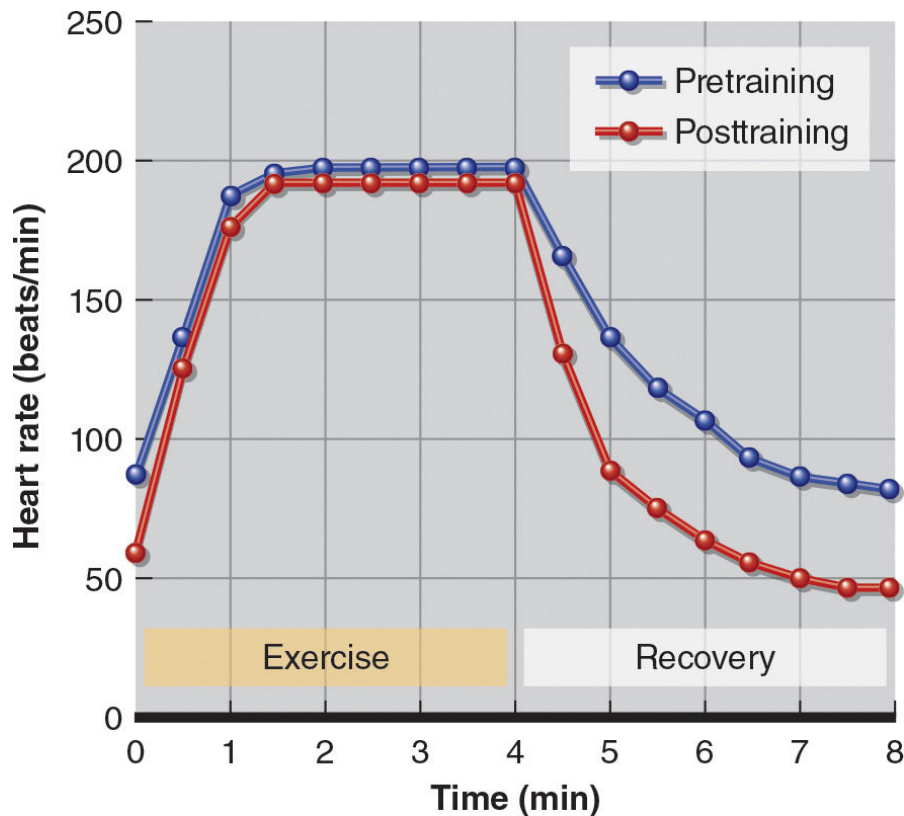


FIGURE 11.5 Changes in heart rate during recovery after a 4 min, all-out bout of exercise before and after endurance training.

Because the heart rate recovery period is shorter after endurance training, this measurement has been proposed as an indirect index of cardiorespiratory fitness. In general, a more fit person recovers faster after a standardized rate of work than a less fit person, so this measure may have some utility in field settings when more direct

measures of endurance capacity are not possible or feasible. However, factors other than training can also affect heart rate recovery time. For example, an elevated core temperature or an enhanced sympathetic nervous system response can prolong heart rate elevation.

The heart rate recovery curve is a useful tool for tracking a person's progress during a training program. But because of the potential influence of other factors, it should not be used to compare individuals.

Cardiac Output

We have looked at the effects of training on the two components of cardiac output: stroke volume and heart rate. While stroke volume increases with training, heart rate generally decreases at rest and during exercise at a given absolute intensity.

Because the magnitude of these reciprocal changes is similar, cardiac output at rest and during submaximal exercise at a given exercise intensity does not change much following endurance training. In fact, cardiac output can decrease slightly. This is likely the result of an increase in the $(a-\bar{v})O_2$ difference (reflecting greater oxygen extraction by the tissues) or a decrease in the rate of oxygen consumption (reflecting an increased mechanical efficiency). Generally, cardiac output matches the oxygen consumption required for any given intensity of effort.

Maximal cardiac output, however, increases considerably in response to aerobic training, as seen in [figure 11.6](#), and is largely responsible for the increase in $\dot{V}O_{2\max}$. This increase in cardiac output must result from an increase in maximal stroke volume, because $HR_{\max}$ changes little, if any. Maximal cardiac output ranges from 14 to 20 L/min in untrained individuals and from 25 to 35 L/min in trained individuals, and can be 40 L/min or more in highly conditioned endurance athletes. These absolute values, however, are greatly influenced by body size.

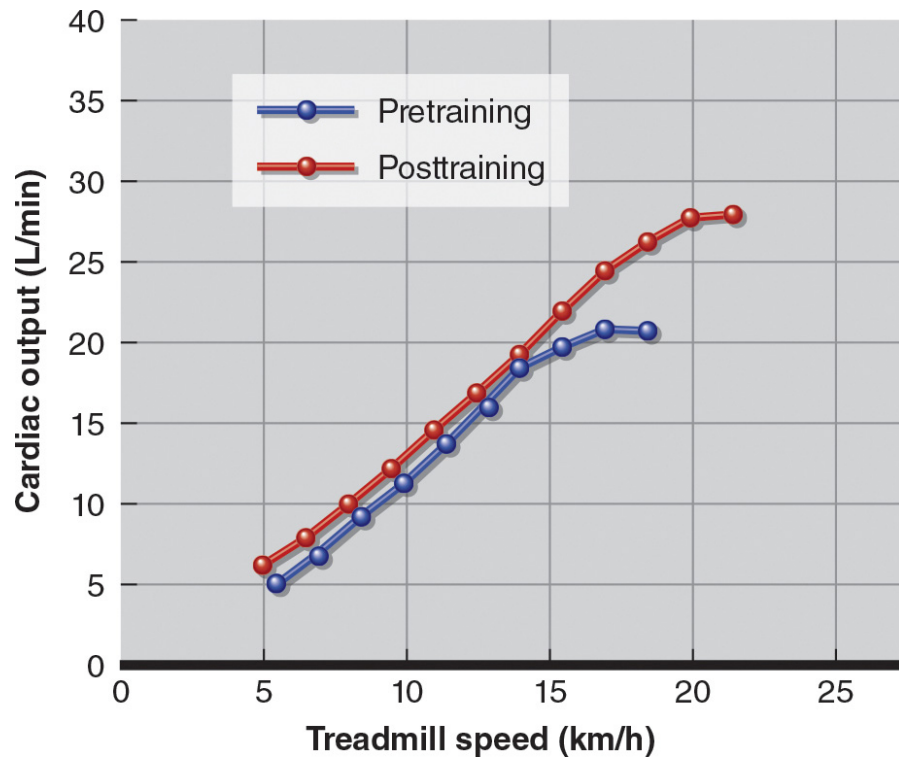


FIGURE 11.6 Changes in cardiac output with endurance training during walking, then jogging, and finally running on a treadmill as velocity increases.

Lundby and colleagues¹⁹ have argued that variability in $\dot{V}O_{2\max}$ among individuals is primarily determined by differences in two variables: maximal cardiac output and red blood cell volume (figure 11.7). (Red cell volume changes are discussed later in this chapter.) Therefore, the response of $\dot{V}O_{2\max}$ to endurance training reflects relative changes in these two important determinants.

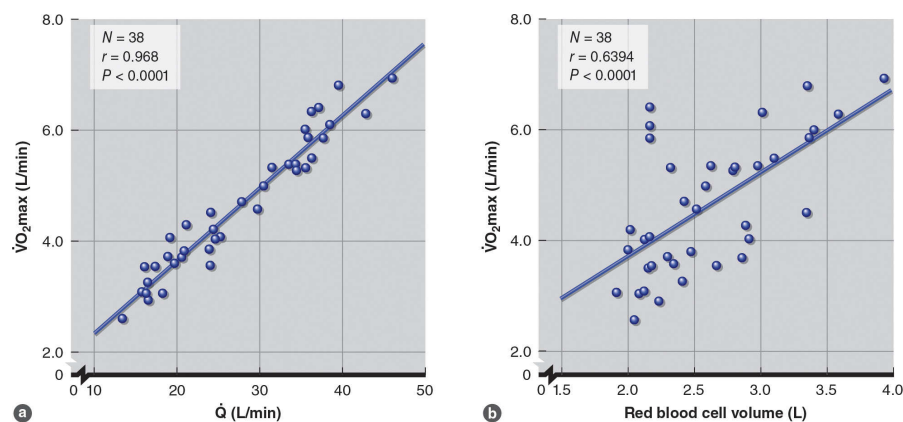


FIGURE 11.7 Correlations between $\dot{V}O_{2\max}$ and (a) maximal cardiac output and (b) red blood cell volume.



In Review

- Resting heart rate decreases as a result of endurance training. In a sedentary person, the decrease is typically about 1 beat/min per week during the initial weeks of training, but smaller decreases have been reported. Highly trained endurance athletes may have resting heart rates of 40 beats/min or lower.
- The mechanisms responsible for the sinus bradycardia associated with endurance training remain controversial, but likely involve both extrinsic (autonomic neural balance) and intrinsic (SA node function) components.
- Heart rate during submaximal exercise is also lower, with larger decreases seen at higher exercise intensities.
- Maximal heart rate either remains unchanged or decreases slightly with training.
- Heart rate during the recovery period decreases more rapidly after training, making it an indirect but convenient way of tracking the adaptations within an individual that occur with training. However, this value is not useful for comparing fitness levels of different people.
- Cardiac output at rest and at submaximal levels of exercise remains unchanged (or may decrease slightly) after endurance training.
- Cardiac output during maximal exercise increases considerably and is largely responsible for the increase in $\dot{V}O_{2\max}$. The increased maximal cardiac output is the result of the substantial increase in maximal stroke volume, made possible by training-induced changes in blood volume and cardiac structure and function.

Blood Flow

Active muscles need substantially more oxygen and fuel substrates than inactive ones. To meet these increased needs, more blood must be delivered to these muscles during exercise. With endurance training, the cardiovascular system adapts to increase blood flow to exercising muscles to meet their higher demand for oxygen and

metabolic substrates. In addition to changes in the heart that allow for better pumping and increased stroke volume, four factors account for this enhanced blood flow to muscle following training:

- Increased capillarization
- Greater recruitment of existing capillaries
- More effective blood flow redistribution away from inactive regions
- Increased total blood volume

To permit increased blood flow, new capillaries develop in trained muscles. This allows the blood flowing into skeletal muscle from arterioles to more fully perfuse the active fibers. This increase in capillaries usually is expressed as an increase in the number of capillaries per muscle fiber, or the **capillary-to-fiber ratio**. Table 11.2 illustrates the differences in capillary-to-fiber ratios between well-trained and untrained men, both before and after exercise.¹⁵

In all tissues, including muscle, not all capillaries are open at any given time. In addition to new capillary formation, existing capillaries in trained muscles can be recruited and open to flow, which also increases blood flow to muscle fibers. The increase in new capillaries with endurance training and increased capillary recruitment combine to increase the overall area for diffusion of oxygen between the vascular system and the metabolically active muscle fibers.

TABLE 11.2 Muscle Fiber Capillarization in Well-Trained and Untrained Men

Stage	Capillaries per mm ²	Muscle fibers per mm ²	Capillary-to-fiber ratio	Diffusion distance ^a
Well trained				
Preexercise	640	440	1.5	20.1
Postexercise	611	414	1.6	20.3
Untrained				
Preexercise	600	557	1.1	20.3
Postexercise	599	576	1.1	20.5

Note. This table illustrates the larger size of the muscle fibers in the well-trained men in that they had fewer fibers for a given area (fibers per mm²). They also had an approximately 50% higher capillary-to-fiber ratio than the untrained men.

^aDiffusion distance is expressed as the average half-distance between capillaries on the cross-sectional view expressed in micrometers.

Adapted by permission from L. Hermansen and M. Wachtlova, "Capillary Density of Skeletal Muscle in Well Trained and Untrained Men," *Journal of Applied Physiology* 30 (1971): 860-863.

A more effective redistribution of cardiac output also can increase blood flow to the active muscles. Blood flow is directed to the active

musculature and shunted away from areas that do not need high flow. Blood flow can increase to the more active fibers even within a specific muscle group. Armstrong and Laughlin² first demonstrated that endurance-trained rats could redistribute blood flow to their most active tissues during exercise better than untrained rats could. The total blood flow to the exercising hindlimbs did not differ between the trained and untrained rats. However, the trained rats distributed more of their blood to the most oxidative muscle fibers, effectively redistributing the blood flow away from the glycolytic muscle fibers. These findings are difficult to replicate in humans because of measurement challenges, as well as the fact that human skeletal muscle is a mosaic with mixed fiber types among individual muscles.

Finally, the body's total blood volume increases with endurance training, providing more blood to meet the body's many blood flow needs during endurance activity without compromising venous return, as discussed next in this chapter.

Blood Volume

Endurance training increases total blood volume, and this effect is larger with higher training intensities. Furthermore, the effect occurs rapidly. This increased blood volume results primarily from an increase in plasma volume, but there is also an increase in the volume of red blood cells. The time course and mechanism for the increase of each of these components of blood are quite different.

Plasma Volume The increase in plasma volume with training is thought to result from two mechanisms. The first mechanism, which has two phases, results in increases in plasma proteins, particularly albumin. Recall from [chapter 8](#) that plasma proteins are the major driver of oncotic pressure in the vasculature. As plasma protein concentration increases, so does oncotic pressure, and fluid is reabsorbed from the interstitial fluid into the blood vessels. During an intense bout of exercise, proteins leave the vascular space and move into the interstitial space. They are then returned in greater amounts through the lymph system. It is likely that the first phase of rapid plasma volume increase is the result of the increased plasma albumin, which is noted within the first hour of recovery from the first training bout. In the second phase, protein synthesis is turned on (upregulated) by

repeated exercise, and new proteins are formed. With the second mechanism, exercise increases the release of antidiuretic hormone and aldosterone, hormones that cause reabsorption of water and sodium in the kidneys, which increases blood plasma. That increased fluid is kept in the vascular space by the oncotic pressure exerted by the proteins. Nearly all of the increase in blood volume during the first 2 weeks of training can be explained by the increase in plasma volume. This early blood volume expansion allows stroke volume to increase despite the fact that changes in the structure and function of the heart itself take longer to develop.

Red Blood Cells An increase in red blood cell volume with endurance training also contributes to the overall increase in blood volume ([figure 11.7b](#)) and red cell volume, like cardiac output, is correlated to $\dot{V}O_{2\max}$. Although the actual number of red blood cells may increase, the hematocrit—the ratio of the red blood cell volume to the total blood volume—may actually decrease. [Figure 11.8](#) illustrates this apparent paradox. Notice that the hematocrit is reduced even though there has been a slight increase in red blood cells. A trained athlete's hematocrit can decrease to such an extent that the athlete appears to be anemic on the basis of a relatively low concentration of red cells and hemoglobin (“pseudoanemia”).

The increased ratio of plasma to cells resulting from a greater increase in the fluid portion reduces the blood's viscosity, or thickness. Reduced viscosity may aid the smooth flow of blood through the blood vessels, particularly through the smaller vessels such as the capillaries. One of the physiological benefits of decreasing blood viscosity is that it enhances oxygen delivery to the active muscle mass.

Both the total amount (absolute values) of hemoglobin and the total number of red blood cells are typically elevated in highly trained athletes. This ensures that the blood has more than adequate oxygen-carrying capacity; that is, the blood's ability to deliver oxygen to exercising muscle is not a limiting factor in exercise. The turnover rate of red blood cells also may be higher with intense training.

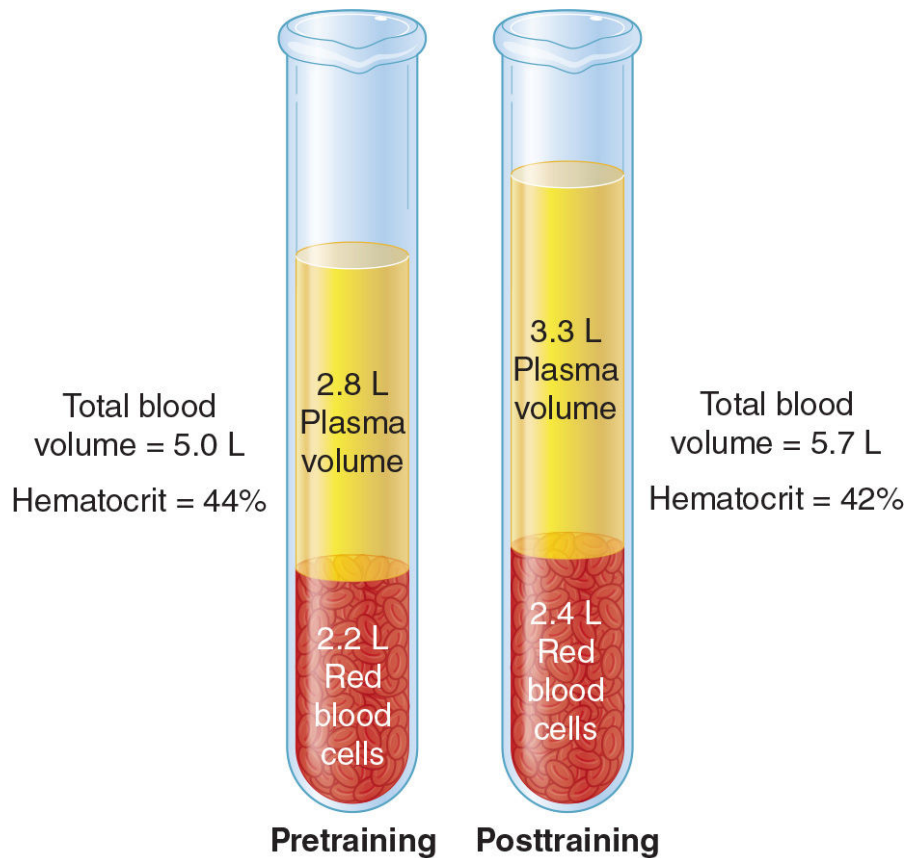


FIGURE 11.8 Increases in total blood volume and plasma volume occur with endurance training. Note that although the hematocrit (percentage of red blood cells) decreased from 44% to 42%, the total volume of red blood cells increased by 10%.



In Review

- Blood flow to active muscle is increased by endurance training.
- Increased muscle blood flow results from four factors:
 1. Increased capillarization
 2. Greater opening of existing capillaries (capillary recruitment)
 3. More effective blood flow distribution
 4. Increased blood volume
- Blood volume increases as a result of endurance training.

- Plasma volume is expanded through increased protein content (returned from lymph and upregulated protein synthesis). This effect is maintained and supported by fluid-conserving hormones.
 - Red blood cell volume also increases, but the increase in plasma volume is typically higher. This decreases blood viscosity, which can improve tissue perfusion and oxygen availability.
-

Respiratory Adaptations to Training

No matter how proficient the cardiovascular system is at supplying blood to exercising muscle, endurance would be hindered if the respiratory system were not able to deliver enough oxygen to fully oxygenate red blood cells. Respiratory system function does not usually limit performance because ventilation can be increased to a much greater extent than cardiovascular function. But, as with the cardiovascular system, the respiratory system undergoes specific adaptations to endurance training to maximize its efficiency.

Pulmonary Ventilation

After training, pulmonary ventilation is essentially unchanged at rest. Although endurance training does not change the structure or basic physiology of the lung, it does decrease ventilation during submaximal exercise by as much as 30% at a given submaximal intensity. Maximal pulmonary ventilation is substantially increased from a rate of about 100 to 120 L/min in untrained sedentary individuals to about 130 to 150 L/min or more following endurance training. Breathing rates typically increase to about 180 L/min in highly trained athletes and can exceed 200 L/min in very large, highly trained endurance athletes. Two factors can account for the increase in maximal pulmonary ventilation following training: increased tidal volume and increased respiratory frequency at maximal exercise.

Ventilation is not usually a limiting factor for endurance exercise performance. However, in some very highly trained athletes, the pulmonary system's capacity for oxygen transport may not be able to meet the demands of exercising muscle and the cardiovascular system. This results in what has been termed *exercise-induced arterial hypoxemia*, in which arterial oxygen saturation decreases below 96%. This desaturation in highly trained elite athletes likely

results from the large right heart cardiac output directed to the lung during exercise and consequently a decrease in the time the blood spends in the lung.

Pulmonary Diffusion

Pulmonary diffusion, or gas exchange occurring in the alveoli, is unaltered at rest and during submaximal exercise following training. However, it increases at maximal exercise intensity. Pulmonary blood flow (blood coming from the right side of the heart to the lungs) increases following training, particularly flow to the upper regions of the lungs when a person is sitting or standing. This increases lung perfusion. More blood is brought into the lungs for gas exchange, and at the same time ventilation increases so that more air is brought into the lungs. This means that more alveoli will be involved in pulmonary diffusion. The net result is that pulmonary diffusion increases.

Arterial–Venous Oxygen Difference

It is clear that stroke volume adapts with endurance training, but peripheral adaptations also contribute to the increase in $\dot{V}O_{2\max}$. The oxygen content of arterial blood changes very little with endurance training. Even though total hemoglobin is increased, the amount of hemoglobin per unit of blood is the same or even slightly reduced. The $(a-\bar{v})O_2$ difference, however, does increase with training, particularly at submaximal exercise intensities. This increase results from a lower mixed venous oxygen content, reflecting both greater oxygen extraction by active tissues and a more effective distribution of blood flow to active tissues. The increased extraction results in part from an increase in oxidative capacity of active muscle fibers, as described later in this chapter.

This was demonstrated in a unique longitudinal study involving both exercise training and a bed-rest deconditioning model.²⁴ Five 20-year-old men were tested (baseline values), placed on bed rest for 20 days (deconditioning), and then trained for 60 days, starting immediately at the conclusion of bed rest. These same five men were restudied 30 years later at the age of 50; they were tested at baseline in a relatively sedentary state and after 6 months of endurance training. The average percentage increases in $\dot{V}O_{2\max}$ were similar for the subjects at age 20 (18%) and at age 50 (14%). However, the

increase in $\dot{V}O_{2\max}$ at age 20 was explained by increases in both maximal cardiac output and maximal $(a-\bar{v})O_2$ difference; at age 50, the increase was explained primarily by an increase in $(a-\bar{v})O_2$ difference, while maximal cardiac output was unchanged. Maximal stroke volume was increased after training at both age 20 and age 50 but to a lesser degree at age 50 (+16 ml/beat at age 20 versus +8 ml/beat at age 50).

While most studies have shown an increase in maximal $(a-\bar{v})O_2$ difference after aerobic training, a 2015 analysis of the literature challenged this long-held notion.²⁵ That study reported that, based on a survey of 13 studies that measured both cardiac output and $(a-\bar{v})O_2$ difference before and after training, improvements in $\dot{V}O_{2\max}$ following 5 to 13 weeks of training were associated with increases in cardiac output, but not in $(a-\bar{v})O_2$ difference. That an increase in maximal cardiac output is the predominant factor associated with increases in $\dot{V}O_{2\max}$ is not surprising, given the close relation between these variables shown in [figure 11.7a](#). However, the training period in the studies analyzed was relatively short, so training adaptations may not have been complete. In longer term endurance training studies, maximal $(a-\bar{v})O_2$ differences were enhanced by 1% to 29%²⁵.

In summary, the respiratory system is quite adept at bringing adequate oxygen into the body. For this reason, the respiratory system seldom limits endurance performance. Not surprisingly, the major training adaptations noted in the respiratory system are apparent mainly during maximal exercise, when all systems are being maximally stressed.

In Review

- Unlike what happens with the cardiovascular system, endurance training has little effect on lung structure and function.
- To support increases in $\dot{V}O_{2\max}$, there is an increase in pulmonary ventilation during maximal effort following training as both tidal volume and respiratory rate increase.
- Pulmonary diffusion at maximal intensity increases, especially to upper regions of the lung that are not normally perfused.
- Although the largest part of the increase in $\dot{V}O_{2\max}$ results from the increases in cardiac output and muscle blood flow, an increase in $(a-\bar{v})O_2$ difference also plays

a key role.

- This increase in $(a-\bar{V}O_2)$ difference is attributable to a more effective distribution of arterial blood away from inactive tissue to the active tissue and an increased ability of active muscle to extract oxygen.
-

Adaptations in Muscle

Repeated excitation and contraction of muscle fibers during endurance training stimulate changes in their structure and function. Our main interest here is in aerobic training and the changes it produces in muscle fiber type, mitochondrial function, and oxidative enzymes.

Muscle Fiber Type

As noted in [chapter 1](#), low- to moderate-intensity aerobic activities rely extensively on type I (slow-twitch) fibers. In response to aerobic training, type I fibers become larger. More specifically, they develop a larger cross-sectional area, although the magnitude of change depends on the intensity and duration of each training bout and the length of the training program. Increases in cross-sectional area of up to 25% have been reported. Fast-twitch (type II) fibers, because they are not being recruited to the same extent during endurance exercise, generally do not increase cross-sectional area.

Most early studies showed no change in the percentage of type I versus type II fibers following aerobic training, but subtle changes were noted among the different type II fiber subtypes. Type IIx fibers have a low oxidative capacity and are recruited less often than type IIa fibers during aerobic exercise. However, during long-duration exercise, these fibers may eventually be recruited to perform in a manner resembling type IIa fibers. This can cause some type IIx fibers to take on the characteristics of the more oxidative type IIa fibers. Recent evidence suggests that not only is there a transition of type IIx to IIa fibers but also there can be a transition of type II to type I fibers. The magnitude of change is generally small, not more than a few percent. As an example, in the HERITAGE Family Study,²⁸ a 20-week program of aerobic training increased type I fibers from 43% pretraining to almost 47% posttraining and decreased type IIx fibers from 20% to 15%, with type IIa remaining essentially unchanged.

These more recent studies have included larger numbers of subjects and have taken advantage of improved measurement technology; both might explain why fiber type composition changes within a muscle are now recognized.

Capillary Supply

One of the most important adaptations to aerobic training is an increase in the number of capillaries surrounding each muscle fiber. [Table 11.2](#) illustrates that endurance-trained men have considerably more capillaries in their leg muscles than sedentary individuals.¹⁵ With long periods of aerobic training, the number of capillaries may increase by more than 15%.²⁸ Having more capillaries allows for greater exchange of gases, heat, nutrients, and metabolic by-products between the blood and contracting muscle fibers. In fact, the increase in capillary density (i.e., increase in capillaries per muscle fiber) is potentially one of the most important alterations in response to training that causes the increase in $\dot{V}O_{2\max}$. It is now clear that the diffusion of oxygen from the capillary to the mitochondria is a major factor limiting the maximal rate of oxygen consumption by the muscle. Increasing capillary density facilitates this diffusion, thus maintaining an environment well suited to energy production and repeated muscle contractions.

Myoglobin Content

When oxygen enters the muscle fiber, it binds to myoglobin, a molecule similar to hemoglobin. This iron-containing molecule shuttles the oxygen molecules from the cell membrane to the mitochondria. Type I fibers contain large quantities of myoglobin, which gives these fibers their red appearance (myoglobin is a pigment that turns red when bound to oxygen). Type II fibers, on the other hand, are highly glycolytic, so they contain (and require) little myoglobin—hence their whiter appearance. More important, their limited myoglobin supply limits their oxidative capacity, resulting in poor endurance for these fibers.

Myoglobin transports oxygen and releases it to the mitochondria when oxygen becomes limited during muscle action. This oxygen reserve is used during the transition from rest to exercise, providing oxygen to the mitochondria during the lag between the beginning of

exercise and the increased cardiovascular delivery of oxygen. Endurance training has been shown to increase muscle myoglobin content by 75% to 80%. This adaptation clearly supports a muscle's increased capacity for oxidative metabolism after training.

Mitochondrial Function

As noted in [chapter 2](#), oxidative energy production takes place in the mitochondria. Not surprisingly, aerobic training also induces changes in mitochondrial function that improve the muscle fibers' capacity to produce ATP. The ability to use oxygen and produce ATP via oxidation depends on the number and size of the muscle mitochondria. Both increase with aerobic training.

During one study that involved endurance training in rats, the number of mitochondria increased approximately 15% during 27 weeks of exercise.¹⁶ Average mitochondrial size also increased by about 35% over that training period. As with other training-induced adaptations, the magnitude of change depends on training volume.

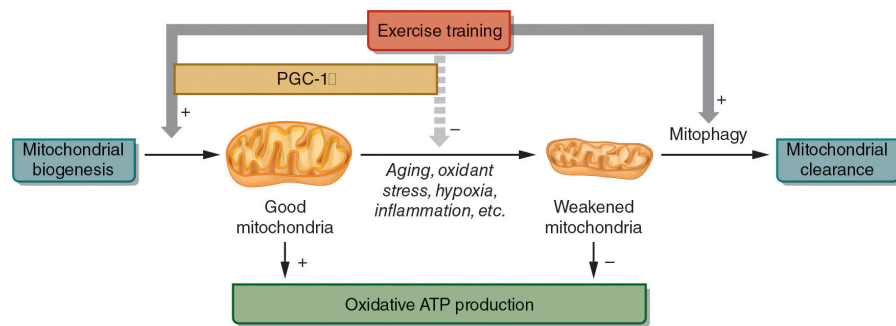


FIGURE 11.9 Endurance exercise training affects the quality of muscle mitochondria by increasing the production of new, healthy mitochondria (biogenesis), decreasing the degradation of mitochondria, and clearing away damaged mitochondria (mitophagy). The first two processes are controlled by the regulator protein PGC-1 α . Solid arrows indicate a positive effect while dotted arrows indicate a negative effect.

Not all mitochondria within a muscle fiber are equally efficient, as new mitochondria are constantly being formed (biogenesis) and old, weakened mitochondria are being cleared (mitophagy) (see [figure 11.9](#)). Regulation of this mitochondrial turnover cycle determines not only the number of mitochondria in a fiber but also the overall quantity and function of those mitochondria,³⁶ which in turn determine overall metabolic function and performance of skeletal muscles. There has been an explosion of new research aimed at understanding the